

Recurrent Sandpile Translations on Finite Sink Graphs

Route-A structural certificate C176

Abstract

For every finite connected undirected loopless sink multigraph and every nonnegative chip vector, we identify addition–stabilization on recurrent stable configurations with translation on the reduced-Laplacian critical group. The order of the translation determines every recurrent orbit, fixed count, Artin–Mazur factor, and finite Koopman determinant.

Keywords: Abelian sandpile; recurrent configuration; critical group; graph Laplacian; dynamical zeta; Koopman permutation.

1 Recurrent representative bridge

Let G be a finite connected undirected loopless multigraph with sink s , non-sink set V° , and reduced Laplacian Δ . Stable heights obey $0 \leq \eta_v < \deg(v)$. For $b \in \mathbb{N}^{V^\circ}$, one clock step is

$$T_b(\eta) = \text{stab}(\eta + b).$$

The sink makes stabilization terminate, and the Abelian least-action argument makes the stable result independent of toppling order.

The critical group is

$$K(G, s) = \mathbb{Z}^{V^\circ} / \Delta \mathbb{Z}^{V^\circ}, \quad D = |K(G, s)| = \det \Delta.$$

The burning criterion selects exactly one recurrent stable representative in every critical-group class. Hence the recurrent set has D states and T_b is translation by $[b]$.

2 Uniform recurrent cycles

Let

$$L = \text{ord}_{K(G, s)}([b]) = \min\{\ell \geq 1 : \ell b \in \Delta \mathbb{Z}^{V^\circ}\}.$$

Theorem 1. *Every recurrent orbit has exact length L , and there are D/L primitive cycles. For every $n \geq 1$,*

$$\#\text{Fix}(T_b^n) = D \mathbf{1}_{L|n}, \quad \zeta_{T_b}(z) = (1 - z^L)^{-D/L},$$

and the finite recurrent Koopman determinant is $\det(I - zU_b) = (1 - z^L)^{D/L}$.

Indeed $T_b^n(\eta) = \eta$ exactly when $n[b] = 0$, independently of η . Division of the D states by the common least period gives the cycle count, and each cycle contributes one reciprocal zeta factor and one determinant factor.

3 Two exact order coordinates

If

$$U\Delta V = \text{diag}(d_1, \dots, d_r)$$

is a Smith decomposition with unimodular U, V , then $\ell[b] = 0$ exactly when $d_i \mid \ell(Ub)_i$ for every i . Thus

$$L = \text{lcm}_i \frac{d_i}{\gcd(d_i, (Ub)_i)}.$$

Alternatively set $w = \text{adj}(\Delta)b$. Since $\Delta^{-1} = \text{adj}(\Delta)/D$,

$$L = \text{lcm}_i \frac{D}{\gcd(D, w_i)} = \frac{D}{\gcd(D, w_1, \dots, w_r)}.$$

Both formulas give $L = 1$ for $b = 0$. If there is no non-sink vertex, the empty determinant convention gives $D = L = 1$ and one empty recurrent state.

4 Characters, reversal, and self-adjointness

Each length- L cycle contributes all L -th roots of unity once. Equivalently, critical-group characters diagonalize translation, and restriction to $\langle [b] \rangle$ has fibres of size D/L . Every L -th root therefore occurs with multiplicity D/L .

Group inversion $R(\eta) = \eta^{-1}$ is involutive and obeys $RT_bR = T_b^{-1}$. Hence $\Theta f = \overline{f \circ R}$ is antiunitary and $\Theta U_b \Theta = U_b^{-1}$. Finally

$$U_b = U_b^* \iff T_b = T_b^{-1} \iff 2[b] = 0 \iff L \leq 2.$$

5 Route-A boundary

The source cycles are intrinsic but have no prime semantics or target divisor comparison. The preliminary v0.2 tuple is

$$(\text{AO_FAIL}, \text{A1_WEAK}, \text{A2_FAIL}, \text{A3_FAIL}, \text{A4_NATURAL_QUANTIZATION}),$$

with overall `ROUTE_A_REJECTED` and Route B false. The positive operator statement is restricted to recurrent configurations; no permutation claim is made for all stable states. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.